

When robots reshape teams: Neurodynamic insights into taskwork and teamwork in search and rescue

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Search and Rescue (SAR) Environments

- Responders are constrained by (Bearman et al., 2023):
 - Spatial demands
 - Dynamic environments
 - Limited ability to accomplish tasks in situ
- Requiring close coordination (Lyons et al., 2017):
 - Shared knowledge
 - Tracking current + past tasks

SAR Requires a Blend of...

- *Taskwork:*
 - Accomplishing individual responsibilities
 - Linked to individual task performance + error management (Salas et al., 2005)
- *Teamwork:*
 - Coordination, collaboration, and cooperation with another agent
 - Linked to collective performance of a team (Cooke et al., 2013)

Taskwork vs. Teamwork

- Teamwork:
 - Team cognition improves team resilience (McNeese et al., 2018)
 - Social features of teams improve context-dependent outcomes (Rahman et al., 2018)
- While many existing works have placed an emphasis on teamwork, the importance of taskwork should not be understated
- Individual abilities (related to taskwork) are sensitive to the demands of SAR
 - Executive-function-related skills are impacted by cognitive workload, fatigue, and stress (Manges et al., 2020)

Robots in SAR

- *Disaster Robotics*: support the demands of first responders
 - Reduce physical demands (Zhang et al., 2025)
 - Reduce cognitive load (Han et al., 2024)
 - Ex. Lifting heavy objects
 - Ex. Navigation

Despite noted benefits, autonomous agents are typically not designed with enough flexibility to support taskwork, and teamwork

Existing Literature on Human Robot Teams

- Has focused on dyadic HRI (1 human, 1 robot)
 - Trust, dependence, safety (Cavicchi et al., 2025)
 - Robots have restricted capabilities
- Human-Robot Teams (HRTs):
 - Less efficient than all human teams in information processing, situation awareness, and performance (Gromek & Lowe, 2025)
 - But can facilitate collaboration and communication between humans (Ouyang & Xu, 2024; Weetathna et al., 2023)
 - Though, many states benefits to the *team* may be reduced to **taskwork** benefits rather than **teamwork** benefits (Ex. reduced workload)

Thus, there is a need to understand how the introduction of a robot impact **both** teamwork and taskwork

Measuring Taskwork

- Several *subjective* measures are used to assess individual states and performance
 - NASA TLX
 - Cognitive Failures Questionnaire
 - Working Memory Questionnaire
- To objectively quantify taskwork, previous work has used used EEG power spectral density (PSD)
 - Theta (4-7 Hz) at Temporo-parietal junction (TPJ) - social cognition (Wang et al., 2016)
 - Theta at prefrontal cortex (PFC) – cognitive workload (Cavanagh & Frank, 2014)

Measuring Teamwork

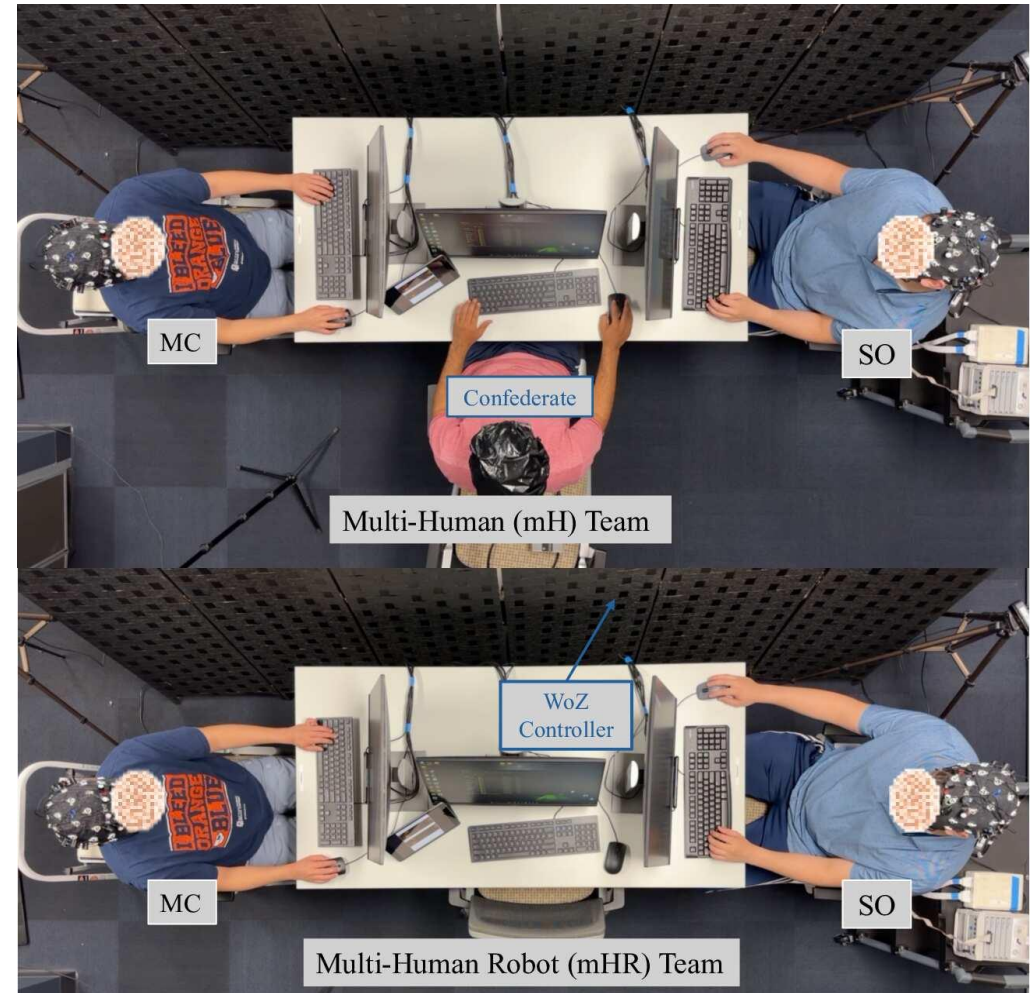
- Existing measures, while comprehensive, are subjective and time consuming
 - TeamSTEPPS
 - Group Environment Questionnaire
- Hyperscanning: measuring interbrain synchrony (IBS) between individuals
 - Cooperating partners have higher IBS than competing partners (Czeszumki et al., 2020)
 - Indication of social cohesion, which may rely on the successful detection of social cues (Hayati et al., 2025)

Hypotheses

- In the present work, we used a combination of PSD and hyperscanning to measure taskwork, and teamwork while working with robots in an SAR environment.
- $H_{1.1}$: Multi-human robot (mHR) teams will reduce the **cognitive load** of highly involved team members
 - Theta PSD at PFC
- $H_{1.2}$: mHR teams will alter **social cognition**
 - Theta PSD at TPJ
- H_2 : Neurodynamic signatures of teams will differ across mHR and multi-human (mH) teams
 - Weighted phase lag index (wPLI)

Method

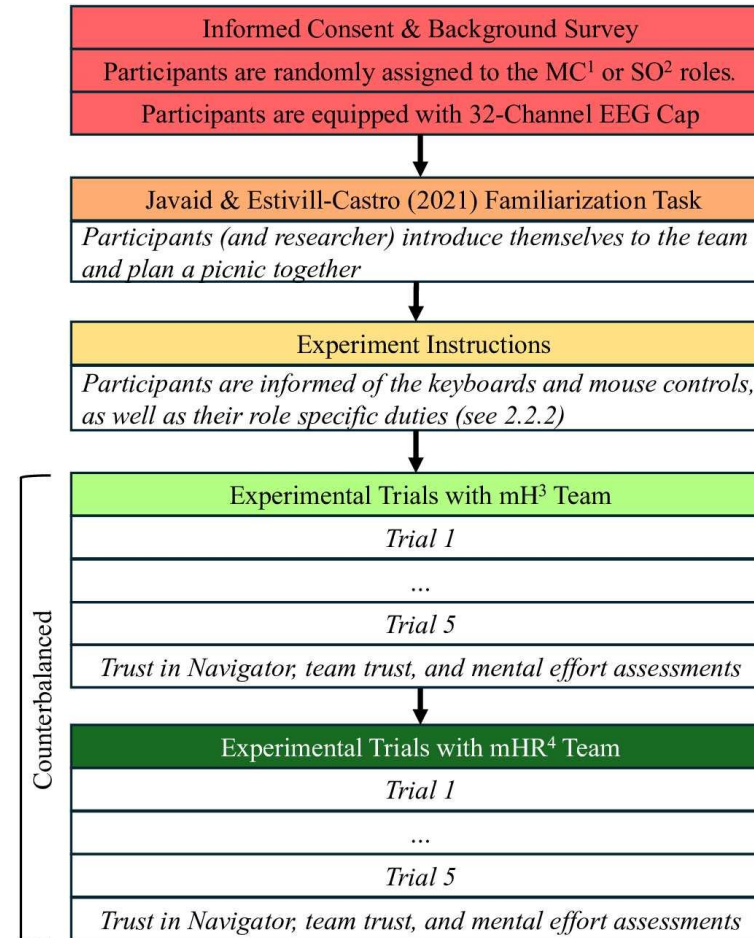
- $n = 42$, forming 21 teams
 - 19 *F*; 23 *M*
 - 4 *F-F*, 7 *M-M*, 11 *F-M*
 - Participants had not met previously
- 32-Channel BrainVision EEG cap
- Randomly assigned to:
 - **Mission commander (MC):** collaborate with team members to gather information + lead team
 - **Safety officer (SO):** monitor nitrous oxide levels, and report danger



Navigator provides information about where victims are located, but only when asked

Experimental Design

- Participants complete a SAR task under 2 team configurations
 - mH (all human)
 - mHR (2 human + 1 WOZ controlled robot)
- Environment built in Unity
 - 2 identical floors
 - 16 randomized victims
 - Random unsafe areas
 - 3-minute trials



Measures

- **Neural**

- Theta PSD (4-7 Hz)
 - Fp1/Fp2
 - TP9/TP10
- Weighted phase lag index (wPLI)
 - Per frequency band
 - Broadband (0.1-30 Hz)

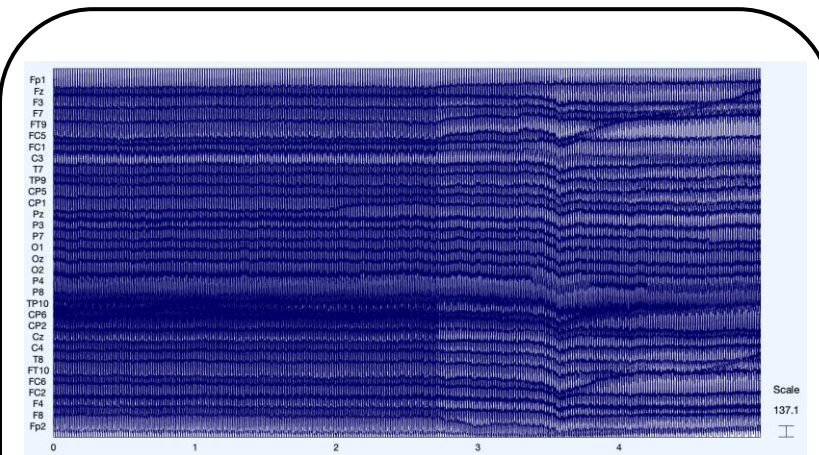
- **Subjective**

- Trust in the navigator
“I trust the navigator” – 7 point
- Mental Effort
“What is your mental effort” – 7 point
- Team trust
“I trust my team” - 7 point

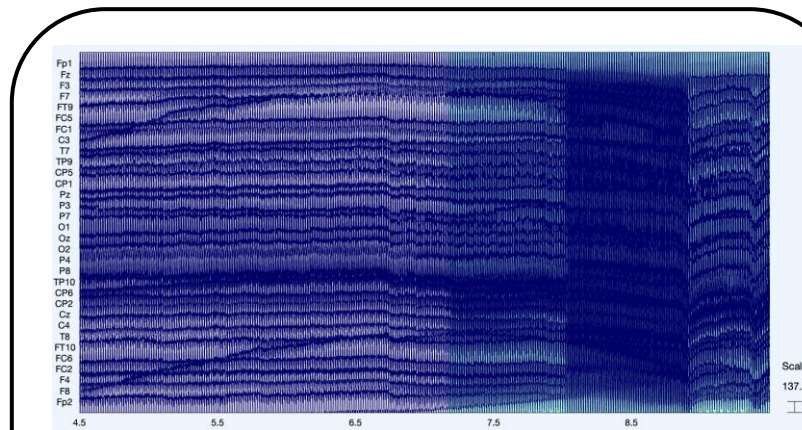
- **Behavioral**

- # of victims located
- Duration team was stopped to communicate

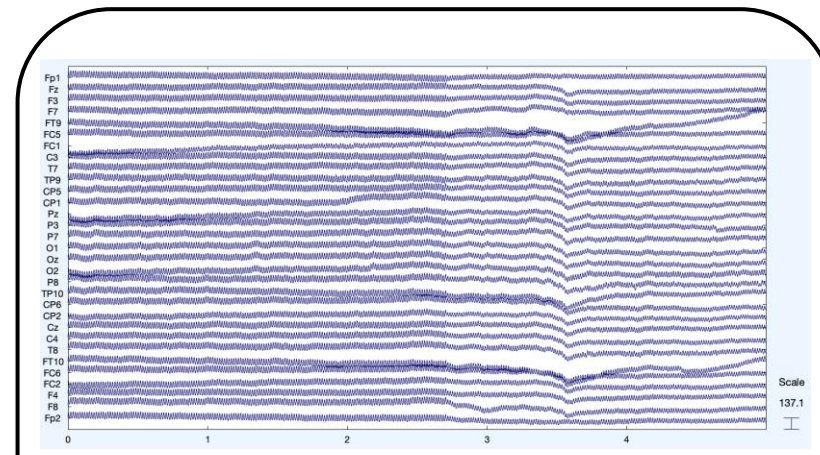
EEG Preprocessing



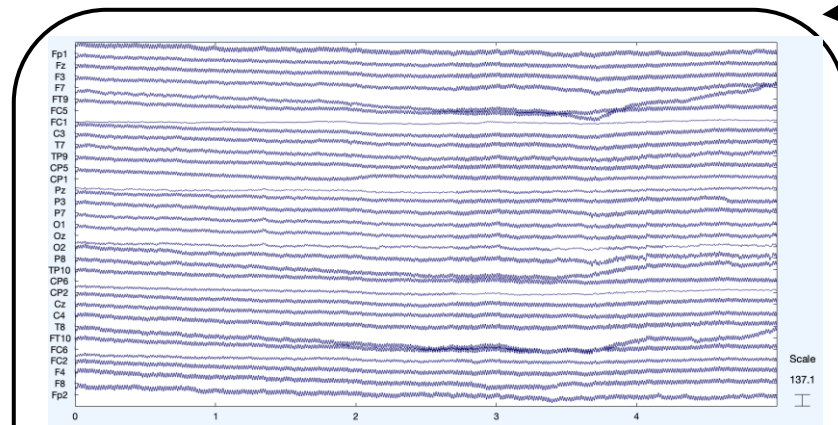
1. Removal of DC offset



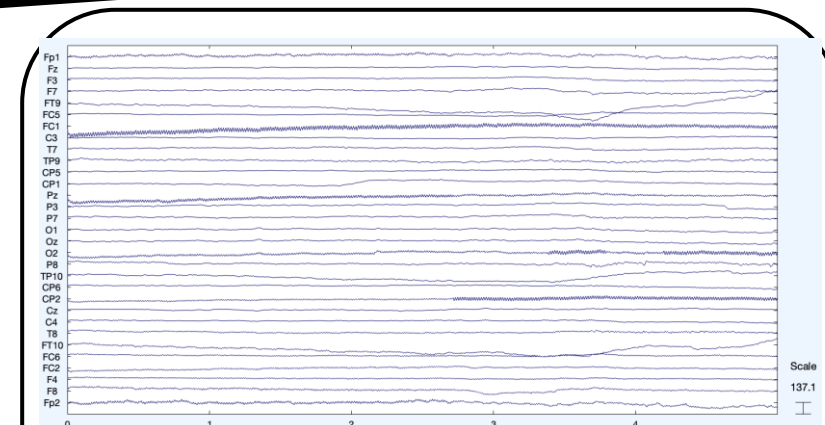
2. Manual rejection of non-neural data



3. 0.1-30 Hz IIR Butterworth filter



4. Independent component analysis (ICA) + ICLabel



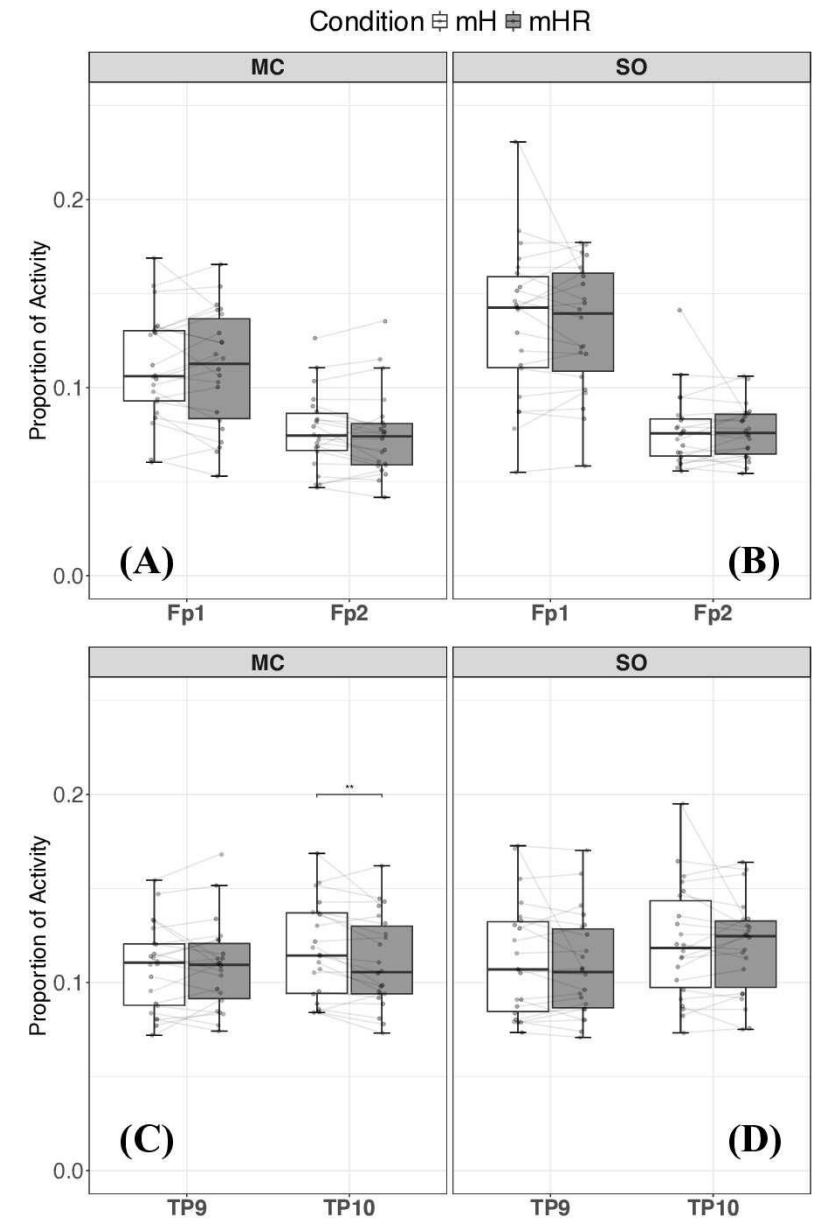
5. Re-referenced to average reference

Neural Measures of Taskwork & Teamwork

- Pre-processed data was partitioned into trials
- Frequency bands were extracted using a 1s sliding window, with a 0.5s Welch's overlap (Bruns, 2004; Klimesch, 1999)
 - Delta (0.5-4 Hz)
 - Theta (4-7 Hz)
 - Beta (8-12 Hz)
 - Alpha (13-30 Hz)
- Weighted Phase Lag Index (wPLI) is then extracted using a Hilbert transform (Vinck et al., 2011)
 - Leader-follow dynamics
 - Reduces the weight of co-activation
- For all neural measures, median value was extracted per trial

Results – Taskwork

- Mission Commanders:
 - Significant difference observed in theta at TP10 ($p = .01$)
 - No other significant differences were observed (all p 's $> .14$)
- Safety Officers:
 - No significant differences were observed across team configurations (all p 's $> .27$)

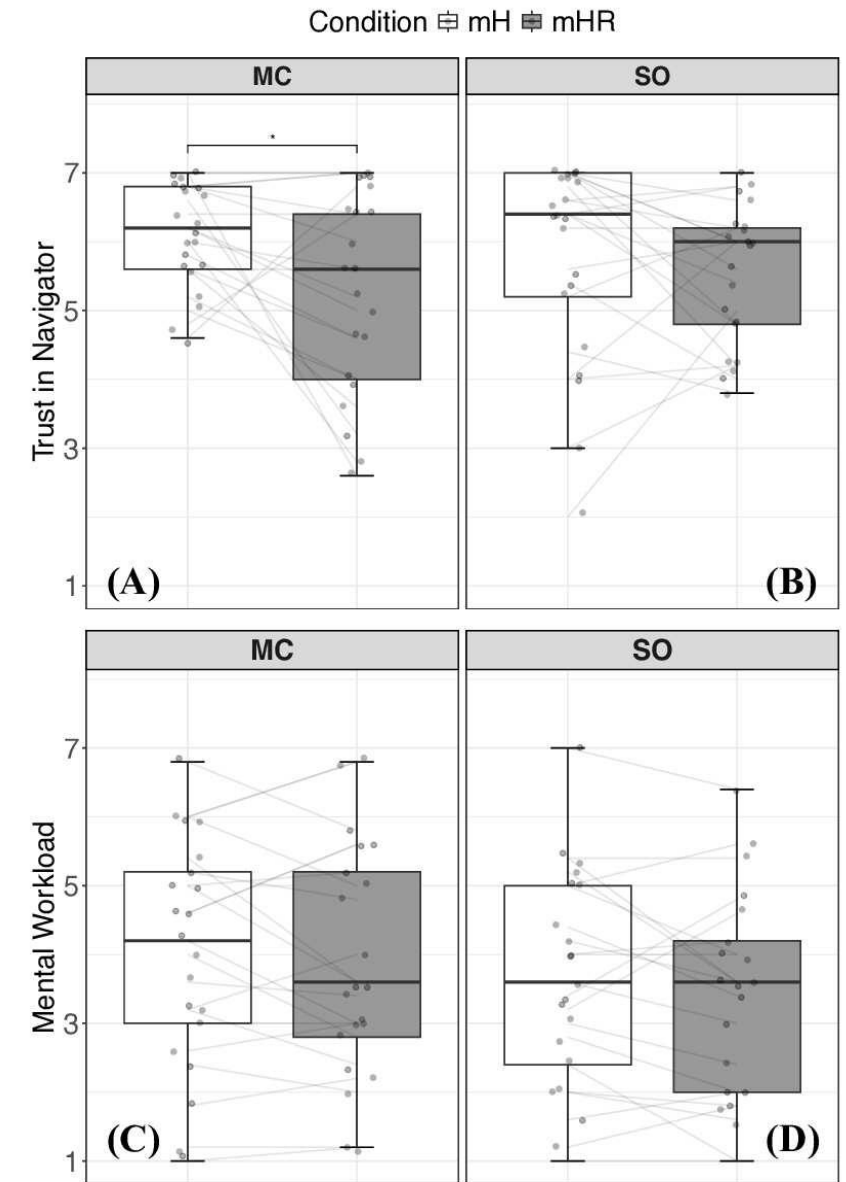


Results – Teamwork

- Significantly higher neural synchrony during mHR team configurations:
 - Alpha – FT9 ($p < .01$)
 - Beta – Fp1 ($p = .05$)
 - Beta – T8 ($p = .04$)
 - Delta – FC2 ($p = .02$)
 - Broadband – FT9 ($p < .01$)
 - Broadband – O1 ($p = .02$)

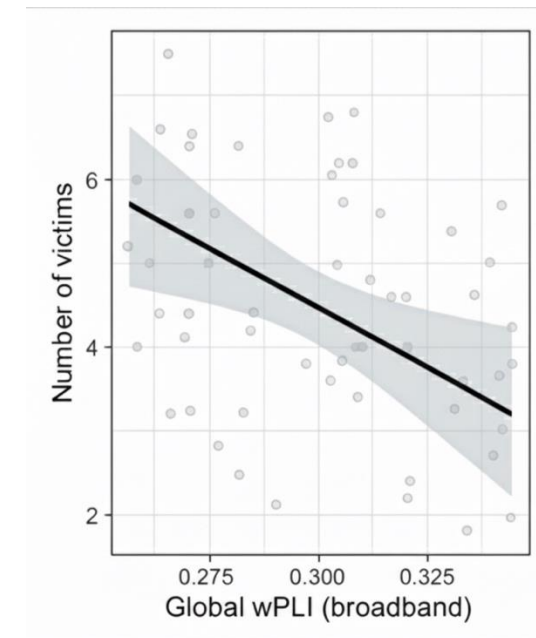
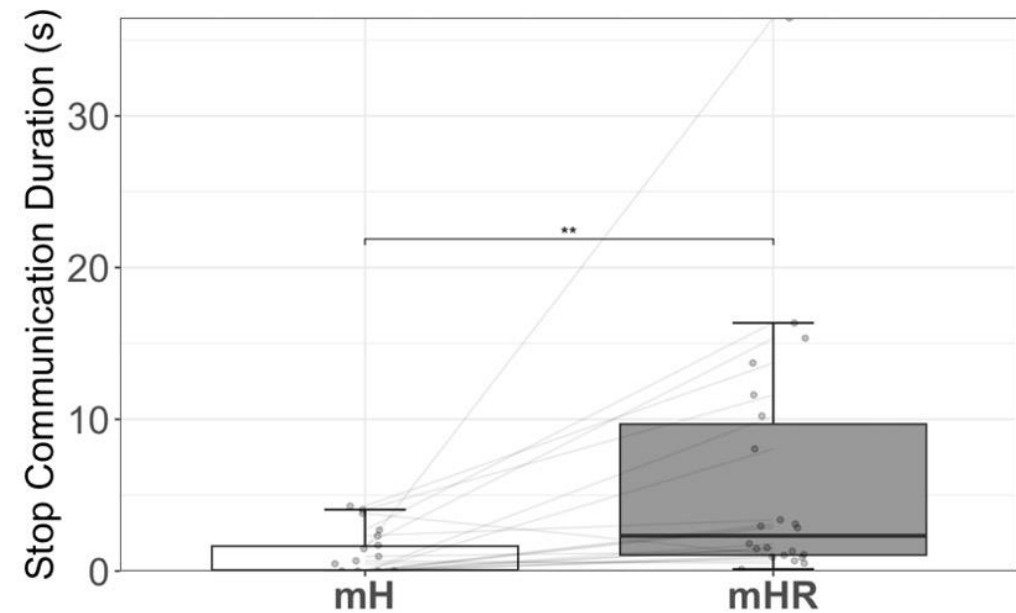
Results – Subjective

- Mission commanders report significantly less **trust in a robot navigator** ($p = .01$), while safety offers do not
- No significant differences were reported in **team trust** or **mental effort** measures (all p 's $> .10$)



Results – Behavioral

- No significant difference in victims located ($p = .18$)
- Teams spend significantly more time stopped to communicate during mHR trials ($p < .01$)
- Global, broadband wPLI negatively predicts team performance ($p = .02$)



Discussion – Taskwork

- $H_{1.1}$ – Not supported
 - Inclusion of a robot did **not** reduce mental workload
 - Contrasts previous work, suggesting that robots will reduce mental effort (Carissoli et al., 2024)
- $H_{1.2}$ – Partially supported
 - Mission commanders were found to have lower TPJ theta, but not safety officers
 - Robot may alter **social cognitive abilities**, particularly of more involved members (Nguyen et al., 2020)
 - Possible that the safety officers had **limited engagement** with the robot

Discussion - Teamwork

- H₂ – supported
 - Left frontotemporal site (**governing social interactions**) synchrony may indicate increased collaboration in mHR teams (Lachat et al., 2012)
 - **May be explained through stopped communication**
 - Increased synchrony at left occipital and prefrontal sites under mHR condition indicates teams engage in more **joint and visual attention** processes (Czeszumski et al., 2022)
 - **May import a “hidden cost” on members**

Limitations

- Exploratory nature
- Median wPLI and PSD values per trial
- Imbalanced sex distribution
- WoZ controller with pre-defined speech options

Conclusions

- A combination of taskwork and teamwork features are altered by the inclusion of a robot, which may be modulated by member involvement
- Neural measures may be more sensitive to these changes compared to subjective measures
- Multi-human robot teams must be viewed as a fundamentally distinct team composition

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Thank you + Questions



Full Paper



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Dr. Aakash Yadav



INTEGRATE

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